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The University of Manchester

Thesis title goes here

A thesis submitted to the University of Manchester for the degree of
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in the Faculty of Science and Engineering

202X

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Abstract

put abstract here

Lay abstract

This is lay abstract text.

Declaration of originality

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Acknowledgements

Acknowledgements go here.

Chapter 1

Example Chapter

1.1 Introduction

Welcome to the unofficial University of Manchester Reproducible Thesis Template. My PhD project was done in an entirely open and reproducible manner, including all papers and my final thesis. I developed this template in case anyone else wanted to have a go at writing a fully reproducible thesis. I am an R user exclusively, so I wrote my thesis using Quarto (https://quarto.org/) and the Rstudio (https://posit.co/download/rstudio-desktop/) Integrated Development Environment, although everything here should also work with VS Code (https://code.visualstudio.com/). In this example/tutorial chapter, I cover:

- The general workflow for using the template.
- How to write using quarto-flavoured markdown.
- How to use citations, cross-referencing, and labels.
- How to use code chunks to generate figures and tables, including how to use pre-rendered graphics.
- How to use code chunks for statistical modelling.
- How to use the scripts provided to render your final document.

1.1.1 A Note for Users of Other Languages

Quarto uses markdown syntax. I use Rstudio as my IDE. `knitr` is the engine that interprets code chunks when using Rstudio, and it supports a variety of languages, including R, Python, SQL, JavaScript, and Ruby. For a full list of supported languages, type the following command:

```
names(knitr::knit_engines$get())
```

1.2 Workflow

This template has been set up to make it as easy as possible to get started writing your fully-reproducible thesis. The general workflow is provided below:

1. Chapters are stored as `.qmd` files in the `chapters_quarto` folder in the parent directory. You should write each chapter separately using these files.

2. Once you are happy with a draft of a chapter, add it to the `% CHAPTERS %` list in `main.tex`. Figure 1.1 shows how to do this, and it must be done for every chapter.
3. Before producing a final thesis version, make sure to edit `main.tex` with your chapters, author information, declarations, abstracts, and acknowledgements.
4. Finally, use the `finalise_thesis.R` script to render each chapter to a `.tex`, reformat the `.tex`, then knit them all together into the final thesis .pdf.

```
%%%%%%%% CHAPTERS %%%%%%%%%%%%%%%%%%%
\chapter{Example Chapter}\label{chap:example_chapter}
\subfile{chapters_tex/chapters_quarto/example_chapter.tex}
```

Figure 1.1. Edit this section in `main.tex` to specify your chapters and chapter labels.

1.3 Writing With Quarto Markdown

1.3.1 Setting Up

To use this template, you should have a few things installed. `tinytex` is what we will use to render the final thesis. `Quarto` is bundled with newer versions of Rstudio, but if you do not have it, you should install it. Install `tinytex` using the following command in the console:

```
install.packages("tinytex")
```

Now that you've installed the tinytex package, you need to install the TinyTex engine itself using the following:

```
tinytex::install_tinytex()
```

The UoM LaTeX thesis template that this template is based on uses LuaTeX, so we need a couple of additional packages to make sure it all runs as intended. Run the following commands in the console:

```
tinytex::tlmgr_install("luatex85") # allows us to use luatex
tinytex::tlmgr_install("tex-gyre") # additional font that we need
```

Each chapter must have a YAML header. This specified options that we need to successfully render each chapter. I recommend using the same YAML header for each chapter, just for convenience After the YAML header, the first thing you should have in each `.qmd` chapter is a `setup` chunk. Each time any code chunk in the document is manually run, the `setup` chunk is automatically run first. In it, you should load in the packages you need, set up any labelling vectors, and deal with function conflicts however you normally do. All you need to do is label the chunk `setup`. The `.qmd` for this tutorial chapter has such a chunk.

1.3.2 The Basics

Quarto.org provides a great guide (https://quarto.org/docs/authoring/markdown-basics.html) to writing with quarto markdown. Ignore the image links, but everything else works as documented. In brief:

italics, **bold**, ***bold italics*** are done like this:

```
*italics*, **bold**, ***bold italics***
```

superscript^2 / subscript$_2$ are done like this:

```
superscript^2^ / subscript~2~
```

`verbatim code` is done like this:

```
`verbatim code`
```

Lists, both numbered and unnumbered, use markdown syntax.

```
- item 1
- item 2
```

Producing:

- item 1
- item 2

And numbered lists:

```
1. item 1
2. item 2
```

Producing:

1. item A
2. item B

Footnotes are fully supported[1].

[1] this is a footnote.

1.3.3 Citations

In this thesis template, numerical citations are used, with a single, long bibliography inserted at the end. `references.bib` is kept in the parent directory. In-text citations are done like this:

`...as found in previous work \cite{strain_2023}...`

producing this text:

…as found in previous work [1]…

1.3.4 Cross-Referencing

Chapters are referenced as such:

`As we can see in \chap{chap:example_chapter}...`, which produces the following:

As we can see in Chapter 1…

Sections are referenced similarly, but you need to type the word "Section" in addition.

`As in, in Section \ref{citations}...` produces the following:

As in, in Section 1.3.3…

Over the whole thesis, you will probably end up with multiple sections called "Introduction", "Methods", "Results" etc. If this is the case, you can manually specify the label for a section like so:

```
# Introduction {#introduction_chapter_2}

# Introduction {#introduction_chapter_3}
```

Which allows you to do the following:

In Section 2.2 as opposed to Section 3.2.

Pages are referenced using `\pageref{}`. This does not automatically include the word "page", so you need to manually add that in. You can point `pageref{}` to any chapter, section, figure, or table like so:

```
page \pageref{chap:example_chapter}   # chapter

page \pageref{introduction_chapter_1} # section

page \pageref{fig-example}            # figure

page \pageref{tbl-example}            # table
```

You have two options for referencing figures and tables. If you are referencing something in the same `.qmd` you are writing in, you can use the `@` symbol like so:

`...as we can see in @fig-example and @tbl-example...`, which produces the following:

...as we can see in Figure 1.2 and Table 1.1...

If you are referencing across different chapters, you need to use native LaTeX referencing style (the same as we use for sections and pages). When doing this, you need to manually add in "Figure" and "Table", so writing this:

`...as we can see in Figure \ref{fig-example} and Table \ref{tbl-example}...` produces the following:

...as we can see in Figure 1.2 and Table 1.1...

1.4 Figures & Tables

For reproducibility, figures and tables should be generated from code using code chunks.

1.4.1 Figures

Quarto offers a number of figure options that you might be familiar with as a LaTeX user. At a minimum, figures require:

- A label that begins with `fig-` and contains no underscores.
- The `include: true` option so that the figure is displayed.
- A figure caption specified by `fig-cap: caption`

I prefer to leave `fig-height` and `fig-width` as `NA` in the YAML, and to instead only specify the `fig-asp` (aspect ratio) in each code chunk. This ensures that figures always take up the full text width in the resulting pdf document.

If you have pre-rendered graphics, such as screenshots of experiments or figures that are taken from other published work, you can use `knitr::include_graphics()` to include them, just as you would in native LaTeX. Below is a code chunk that accomplishes this. `eval` has been set to false so that the figure itself is not displayed.

```
knitr::include_graphics(path = "../supplied_graphics/chapters_tex_example.png", dpi = NA)
```

Code chunks can be inserted using a button in the top right corner of the editing window, or by using `Ctrl+Alt+I`. Have a look at `example_chapter.qmd` in `quarto_chapters` for a demo of this. Figure 1.2 is produced using the R code chunk directly above it.

```r
# generate some numbers, make a dataframe

x <- rep(1:10)
y <- runif(10)

df <- tibble(x, y)

# load plotting library, make basic plot

df %>%
  ggplot(aes(x = x, y = y)) +
  geom_smooth(span = 0.5, se = FALSE)
```

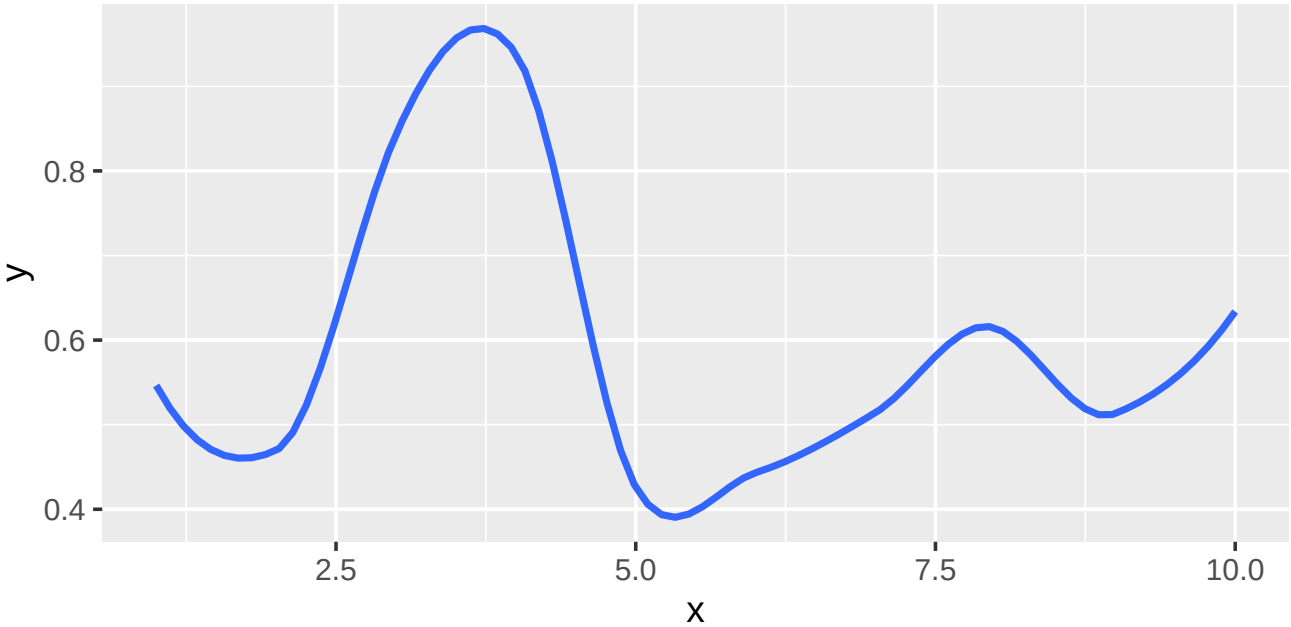

Figure 1.2. An example of a figure.

The YAML for this chapter includes the following option `crop: true`. This instructs `knitr` to use `pdfcrop` to crop produced pdf images. This requires the `pdfcrop` package, which is not bundled with `tinytex` by default. Install it with `tinytex::tlmgr_install("pdfcrop")`. It also requires a `perl` environment. On Windows, I use Strawberry Perl (https://strawberryperl.com/), which is easy to install and set up.

1.4.2 Tables

Most tables you create will by dynamically generated from models/other stats. Otherwise, I prefer to manually create a table in a .csv file, store it in the directory, then read it in and use the `kableExtra` package to format it properly. This is much quicker and more reproducible than manually creating markdown/LaTeX tables. As with figures, chunks that produce tables have several requirements:

- A label that begins with `tbl-` and contains no underscores.

- The `include = TRUE` options so that the figure is displayed.
- A figure caption specified by `tbl-cap: caption`

Below is a code chunk and the corresponding table that it produces:

```
# first, read in the csv that the table data lives in

table_df <- read_csv("../data/table_data.csv") # I use read_csv from the readr package

# next, use kableExtra to format and produce the table

kbl(table_df, booktabs = TRUE)                 # imo booktabs produces the best looking tables
```

Table 1.1. An example of a properly formatted table.

Column 1	Column 2	Column 3
A	1	You
B	2	and
C	3	Me

`kableExtra` also includes options for different kinds of LaTeX styling. See the awesome_table_in_pdf guide online (https://cloud.r-project.org/web/packages/kableExtra/vignettes/awesome_table_in_pdf.pdf) for a full `kableExtra` tutorial.

1.5 Models & Caching

Knitr includes the ability to cache models. This is very useful if your models are especially complex or you have a very large report. This is demonstrated below.

```
set.seed(123)                              # Set seed for reproducibility
x <- 1:100                                 # Predictor variable
y <- 2 * x + rnorm(100, mean = 0, sd = 10) # Response variable with noise

data <- data_frame(x, y)                   # turn into a dataframe

model <- lm(y ~ x, data = data)            # Fit a linear model

summary(model)                             # Summary of the model
```

Setting `eval_models` to `TRUE` in the YAML header means models are evaluated and cached with every knit. Setting to `FALSE` means that knitr looks for the model in the cache, so you only have to build it once! I use the `qwraps2` package to lazyload models in at the start of each chapter that includes modelling. This example chapter does this (see the `.qmd`), and it is reproduced verbatim below:

```
# load in cached models

if (!params$eval_models){ lazyload_cache_dir("../thesis_cache/") }
```

You should do this for every separate chapter that requires cached models.

1.6 Rendering Your Thesis

This repository includes two `.R` scripts that you can use to render your thesis. You only need to interact with `finalise_thesis.R`. Open `finalise_thesis.R` and read the comments there. There are lines of code there to render all chapters (line 6) or render a single chapter (helpful while writing the thesis, line 10). After that, use the script to call `reformat_tex.R` (line 15), then use lines 17 and 18 to use the reformatting script. Once that has all run successfully, you can use line 24 to render the final thesis pdf.

1.7 What's Next?

To ensure full computational reproducibility, we need to make sure that people who want to reproduce the thesis in the future can do so using the same package versions and computational environment. The best way to do this is using a combination of the `renv` package and Docker.

`renv` is an R package that allows you to capture packages, versions, and dependencies in your project. Docker is a piece of software that allows you to package your project so that people can run it virtually in the future.

1.7.1 Capturing Dependencies Using `renv`

Once you're happy with your thesis, install the `renv` package using:

```
install.package("renv")
```

Once you've done that, run the following command in the console:

```
renv::init()
```

This will create several files and folders. The important one is the `renv.lock` file, which will be created in the parent directory. This file captures all packages, versions, and dependencies and makes a snapshot of them. Once you have the `renv.lock` file, you are ready to move on.

1.7.2 Using Docker to Create an Image

First, install Docker desktop (https://www.docker.com/products/docker-desktop/).

Then, write your Dockerfile. For your thesis, the easiest thing to do is to use the Dockerfile that is included in this template repository. Make sure to change the version of R you are using (line 2), and the files that are added to the container (lines 22-36). Lines 39-64 are additional LaTeX packages that I found I needed to knit my thesis together within the Docker container. You can leave everything else as-is, including the `renv` relevant commands (lines 8-19).

1.8 Closing Remarks

I am not going to pretend that writing a fully-reproducible thesis is as easy as using something like Overleaf. If you've made it this far, I am assuming that you are at least motivated to try, good luck! There are idiosyncrasies to writing like this that are best dealt with as they are encountered. My own thesis was produced using this template, and for reference can be found on GitHub (add final link).

Right now, the biggest issue with this is the lack of convenient real-time collaborative editing. I tried `trackdown`, but it is fairly clunky. There are probably other, better ways to do this, but in the end I just sent pdfs to my supervisors and they commented on them the normal way. While it isn't ideal, it worked well enough for me.

Special thanks go Alex Casson, who wrote the original University of Manchester PhD thesis LaTeX template that you can find on Overleaf (https://www.overleaf.com/edu/theuniversityofmanchester#templates). Since then, modifications to the LaTeX `.cls` file have been made by David Petrescu and myself.

References

[1] Gabriel Strain, Andrew J. Stewart, Paul Warren, and Caroline Jay. "The Effects of Contrast on Correlation Perception in Scatterplots". In: *International Journal of Human-Computer Studies* 176 (2023), p. 103040. ISSN: 1071-5819. DOI: 10.1016/j.ijhcs.2023.103040. (Visited on 04/11/2023) (cited on p. 15).

Appendices

Appendix A

First appendix

A.1 Section in Appendix

This is an appendix